

Curves & Bézier Curves - Master Lecture Notes

■ INTRODUCTION TO CURVES

In computer graphics, curves are used to model smooth shapes and paths. There are different ways to

1. Explicit: $y = f(x)$ — limited to single-valued functions.
2. Implicit: $f(x, y) = 0$ — suitable for closed curves (e.g., circles).
3. Parametric: $x = x(t)$, $y = y(t)$ — best for general curve drawing.

■ PARAMETRIC CURVES

Parametric curves use a parameter t (typically from 0 to 1) to define points along the curve. For example $x(t) = \cos(t)$, $y(t) = \sin(t)$ — describes a circle.

■ CONTINUITY IN CURVES

Continuity describes how smooth a curve is at the joint between two segments:

- **C⁰**: Curves connect (same position).
- **C¹**: Tangents match (smooth direction).
- **C²**: Curvatures match (smooth acceleration).

C¹ ensures no sharp turns, C² ensures no sudden change in bending.

■ BÉZIER CURVES

A cubic Bézier curve has 4 control points: P₀, P₁, P₂, P₃

Formula:

$$B(t) = (1 - t)^3 P_0 + 3(1 - t)^2 t P_1 + 3(1 - t)t^2 P_2 + t^3 P_3$$

Key properties:

- Curve starts at P₀, ends at P₃.
- Tangent at start: $3(P_1 - P_0)$, tangent at end: $3(P_3 - P_2)$
- Entire curve lies within convex hull of the control points.

■ DERIVATIVES OF BÉZIER CURVES

1st derivative B'(t): gives the tangent direction.

2nd derivative B''(t): gives the acceleration/curvature.

At endpoints:

$$B''(0) = 6(P_0 - 2P_1 + P_2)$$

$$B''(1) = 6(P_3 - 2P_2 + P_1)$$

■ JOINING TWO CUBIC BÉZIER CURVES

For smooth joining of two Bézier curves A and B: